

PUERTO MONTT, Chile

WMO: **857990**Lat: **41.4350S**Lon: **73.0978W**Elev: **276**StdP: **14.55**Time Zone: **-4.00 (CHL)**Period: **94-19**WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.7	30.5	26.8	21.0	32.8	28.9	23.2	32.7	27.9	48.8	25.3	49.1	1.0	300	0.585

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.6	73.9	62.3	71.4	61.3	68.4	59.7	64.1	71.3	62.5	69.2	60.9	66.6	9.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.1	81.1	66.5	59.3	76.2	64.7	58.2	73.0	63.4	29.4	71.5	28.2	69.5	27.1	66.5	72.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.9	19.7	17.2	DB	25.4	82.7	1.7	3.8	24.1	85.4	23.1	87.6	22.2	89.7	20.9	92.5	(4)
(5)			WB	25.0	67.4	1.8	2.0	23.7	68.9	22.6	70.0	21.6	71.1	20.2	72.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.6	57.8	57.6	54.9	50.8	48.4	45.0	43.9	45.2	46.8	49.6	52.4	55.8
	DBStd	6.39	3.91	4.26	4.06	4.48	4.82	4.82	4.66	4.23	3.94	3.59	3.77	3.91
	HDD50	841	1	1	6	43	89	162	195	159	111	52	18	4
	HDD65	5247	226	209	314	426	514	601	654	615	547	478	378	285
	CDD50	1074	242	215	157	67	40	11	7	8	14	39	90	184
	CDD65	5	2	3	0	0	0	0	0	0	0	0	0	0
	CDH74	122	36	58	10	0	0	0	0	0	0	0	3	15
	CDH80	20	8	10	1	0	0	0	0	0	0	0	0	1
Wind	WSAvg	6.8	6.7	6.6	5.9	6.0	6.6	7.5	7.5	7.8	6.6	6.8	6.8	6.8
Precipitation	PrecAvg	66.9	3.2	3.2	3.7	5.5	8.0	8.4	8.0	7.8	5.2	5.0	4.2	4.1
	PrecMax	90.0	9.7	9.9	7.6	10.6	17.4	14.3	16.5	14.9	9.7	11.3	9.3	12.2
	PrecMin	38.0	0.4	0.4	0.7	2.0	1.2	1.8	3.1	2.9	1.7	0.9	1.1	0.4
	PrecStd	10.2	2.0	2.0	1.5	2.4	3.7	2.8	3.0	2.7	1.9	2.2	1.8	2.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.3	80.9	74.8	69.6	63.7	57.6	57.1	59.3	64.2	68.0	71.2	75.5
		MCWB	65.1	63.0	62.8	60.6	57.3	54.0	53.3	52.4	53.7	56.5	58.6	62.4
	2%	DB	73.2	74.9	69.6	64.3	60.4	55.6	55.0	55.8	59.4	62.8	66.1	71.3
		MCWB	62.5	62.7	60.6	57.8	56.0	52.7	51.6	51.3	51.4	54.1	57.7	60.7
	5%	DB	70.0	71.2	66.3	61.1	57.8	53.9	53.3	53.9	56.9	59.6	63.0	68.1
		MCWB	60.9	62.1	58.9	56.6	54.5	51.4	50.5	50.4	50.8	52.8	55.5	59.1
	10%	DB	67.6	67.8	64.1	59.0	55.8	52.2	51.6	52.1	54.0	57.4	60.7	65.1
		MCWB	59.3	59.8	57.6	55.2	53.2	50.1	49.2	49.0	49.4	51.6	54.1	57.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.6	67.2	64.7	61.9	58.9	55.7	54.8	55.4	55.6	57.7	60.9	64.4
		MCDB	76.7	75.8	70.8	65.6	61.7	57.1	56.0	57.2	61.2	63.8	67.7	72.8
	2%	WB	63.7	64.4	62.0	59.3	57.0	53.7	52.7	53.1	53.3	55.7	58.6	61.8
		MCDB	71.0	71.7	67.3	63.0	59.4	55.2	54.2	55.0	57.4	60.7	64.2	69.2
	5%	WB	61.9	62.5	60.1	57.4	55.3	52.3	51.1	51.3	51.9	54.0	56.8	60.0
		MCDB	68.5	69.3	64.6	60.5	57.7	53.8	52.6	53.4	55.3	58.5	62.1	66.3
	10%	WB	60.1	60.5	58.6	55.7	53.9	50.9	49.6	49.9	50.5	52.5	55.0	58.3
		MCDB	65.8	66.4	62.7	58.5	56.0	52.3	51.0	51.6	53.3	56.5	59.5	63.9
Mean Daily Temperature Range	5% DB	MDBR	18.6	19.0	17.9	15.7	13.3	11.7	12.4	13.1	15.4	15.8	16.4	17.6
		MCDBR	22.9	25.1	22.2	18.5	15.0	11.5	12.5	14.3	19.3	19.8	20.4	22.3
		MCWBR	13.4	14.7	14.1	13.0	11.2	9.4	9.6	10.3	12.5	12.6	12.7	13.2
	5% WB	MCDBR	20.8	22.3	19.1	16.4	12.8	9.8	9.8	10.9	15.7	17.2	18.6	20.3
		MCWBR	13.0	14.0	12.9	12.2	10.1	8.7	8.3	8.7	11.0	11.8	12.5	12.8
Clear-Sky Solar Irradiance	taub	0.332	0.343	0.328	0.320	0.319	0.318	0.315	0.325	0.329	0.327	0.325	0.329	
	taud	2.537	2.500	2.549	2.554	2.539	2.529	2.537	2.508	2.478	2.506	2.533	2.536	
	Ebn at Noon	316	304	295	275	252	240	250	268	289	306	316	318	
	Edh at Noon	34	34	29	25	22	20	21	26	31	33	34	35	
All-Sky Solar Radiation	RadAvg	2160	1854	1351	859	521	377	461	663	1055	1415	1799	2066	
	RadStd	167	141	121	78	71	53	40	73	125	151	169	268	